

Section 12.2: Graphs and surfaces

Definitions. Graphs and cross-sections

Definition: graph of a function of two variables

The graph of a function f is the set of all points (x, y, z) such that

_____.

In general, this graph is a _____ in 3-space.

Definition: cross-section

Holding x fixed at $x = a$ gives the one-variable function _____. Its graph is the curve obtained by intersecting the graph of f with the vertical plane _____. The definition with $y = b$ is analogous.

Example. Describe the graph from its cross-sections

Describe the graph of

$f(x, y) = 1 + x - y.$

Work with cross-sections first; use them to predict the full surface.

Fixed variable	Cross-section equation	Direction of change
$x = a$	$z =$ _____	As y increases, z _____.
$y = b$	$z =$ _____	As x increases, z _____.

Prediction for the full graph: _____

Conclusion from the cross-sections

The graph is a _____; it slopes _____ in the y -direction and _____ in the x -direction.

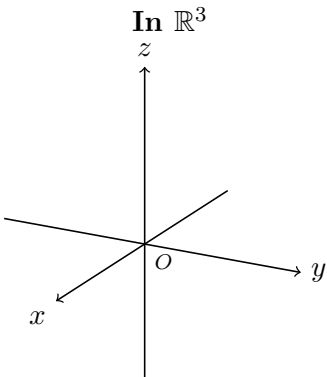
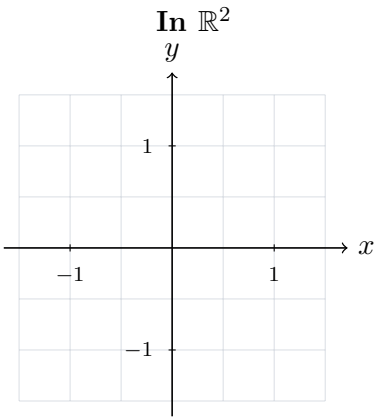
Common surfaces

For each equation, name the surface and decide whether the whole surface is the graph of a function $z = f(x, y)$.

Equation	Surface	Graph $z = f(x, y)$?
$z = 2 - x + 3y$	_____	_____
$z = x^2 + y^2$	_____	_____
$x^2 + y^2 + z^2 = 4$	_____	_____
$z = x^2$	_____	_____

Example. The same equation in two ambient spaces

Graph $x^2 + y^2 = 1$ first in $\mathbb{R}^2$, then in $\mathbb{R}^3$.



Cylinder test

If one variable is missing from an equation in 3-space, the corresponding surface extends parallel to the _____-axis. Here the missing variable is _____, so the surface is a _____.

Problem. Draw cross-sections

Let

$$f(x,y) = y^3 + xy.$$

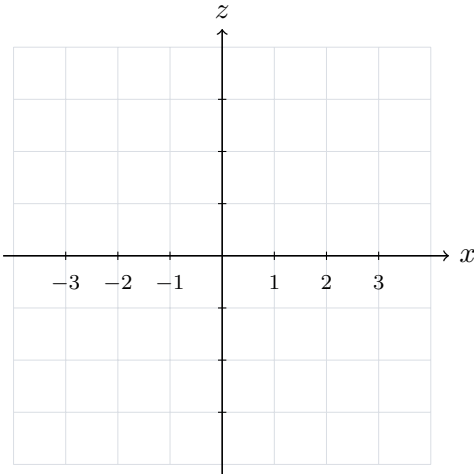
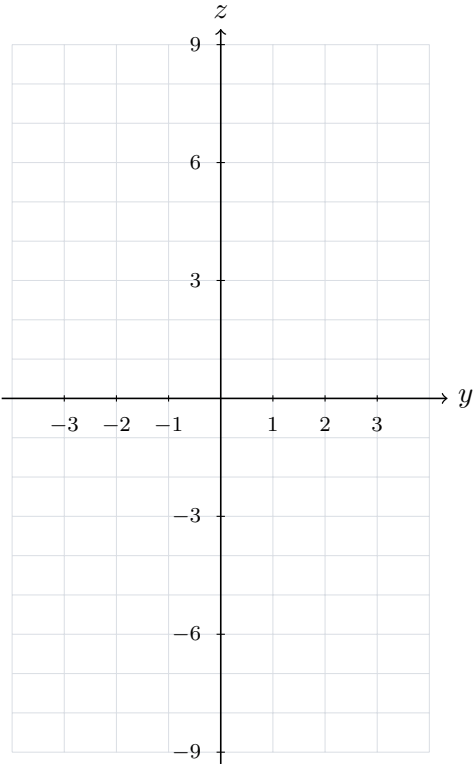
Find the indicated equations, then draw graphs of each family of cross-sections on the same plane.

(a) Fix x .

Value	Cross-section
$x = -1$	$z =$ _____
$x = 0$	$z =$ _____
$x = 1$	$z =$ _____

(b) Fix y .

Value	Cross-section
$y = -1$	$z =$ _____
$y = 0$	$z =$ _____
$y = 1$	$z =$ _____



Section 12.3: Contour diagrams

Recap. Cross-section or contour?

Cross-section

Fix an *input*:

$$x = a \quad \text{or} \quad y = b.$$

Intersect the surface with a _____ plane. The resulting one-variable graph is

$$x = a : \quad z = \underline{\hspace{2cm}},$$
$$y = b : \quad z = \underline{\hspace{2cm}}.$$

Contour or level curve

Fix the *output* at $z = c$. Intersect the surface with a _____ plane, then record the curve in the xy -plane:

_____.

A contour diagram is a collection of level curves labeled with function values.

Example. Contours of a paraboloid

Find equations for the contours of

$$f(x,y) = x^2 + y^2,$$

draw a contour diagram, and relate it to the graph.

General level c :

_____.

c	Contour	Radius
0	_____	_____
2	_____	_____
4	_____	_____
6	_____	_____
8	_____	_____

Contours exist only when

_____.

Relate spacing to the graph

As we move away from the origin, equally spaced levels become _____ together. Therefore, the bowl becomes _____.

Reading contour diagrams

- Distinct contours for different function values cannot _____.
- For equally spaced function values, closer contours indicate _____ change and a _____ graph.
- The numerical labels on the contours are essential.

Example. Compare a cone with the paraboloid

Draw a contour diagram for

$$f(x,y) = \sqrt{x^2 + y^2}$$

and relate it to the graph.

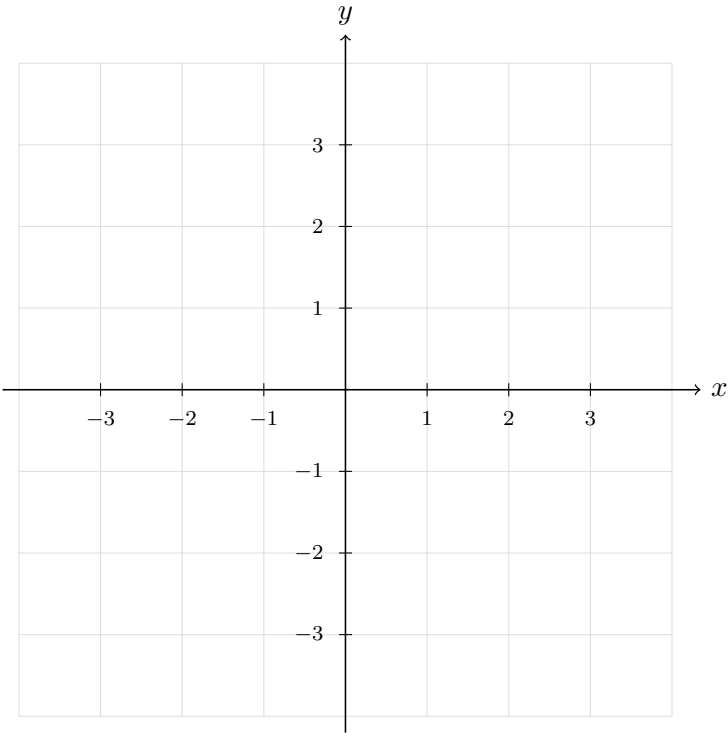
General level c :

_____.

c	Radius
0	_____
1	_____
2	_____
3	_____

The graph is a

_____.



Comparison with the previous example

For the cone, consecutive integer levels give _____ spaced circles, which reflects a _____ slope. For the paraboloid, the circles become _____ together.

Exercise. Linear function

Make a contour plot for

$$z = 2y - x$$

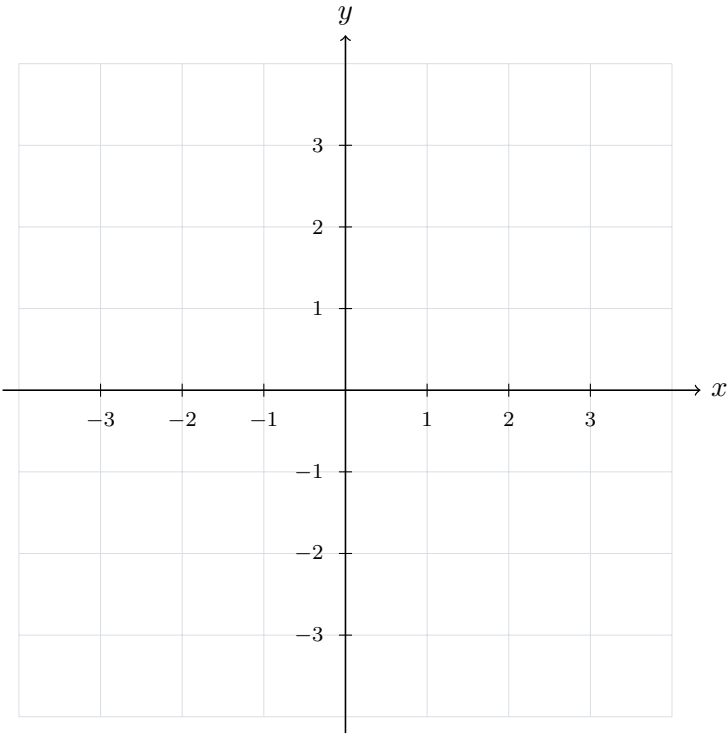
with at least three different contours. Label each contour with its value of z .

For a level $z = c$,

_____.

c	Equation in the xy -plane
0	_____
2	_____
4	_____
6	_____

All contours have slope _____, so
they are _____.



Check the labels

Which direction in the xy -plane gives increasing values of z ?

Example. Build contours from a table

Relate the table for $f(x, y) = x^2 - y^2$ to its contour diagram.

$y \backslash x$	-3	-2	-1	0	1	2	3
3	0	-5	-8	-9	-8	-5	0
2	5	0	-3	-4	-3	0	5
1	8	3	0	-1	0	3	8
0	9	4	1	0	1	4	9
-1	8	3	0	-1	0	3	8
-2	5	0	-3	-4	-3	0	5
-3	0	-5	-8	-9	-8	-5	0

1. Locate the zero entries. Algebraically,

$$x^2 - y^2 = 0 \quad \Longleftrightarrow \quad \underline{\hspace{2cm}}.$$

Thus the $z = 0$ contour consists of _____.

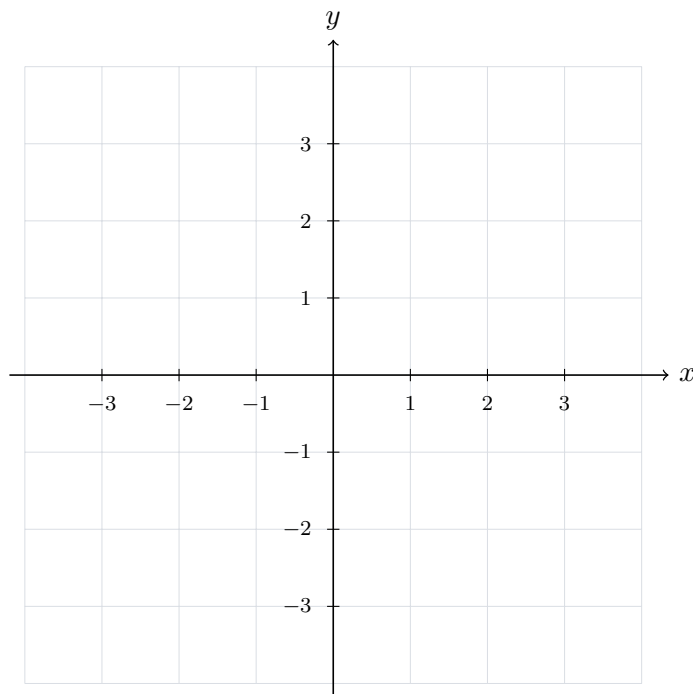
2. Mark the regions in the table where f is positive and where f is negative. What symmetries do you see?

Example, continued. Complete and interpret the contour diagram

On the grid, draw and label the contours

$$z = -4, -2, 0, 2, 4.$$

Use the table to decide where each branch lies.

**Relate the contours to the saddle graph**

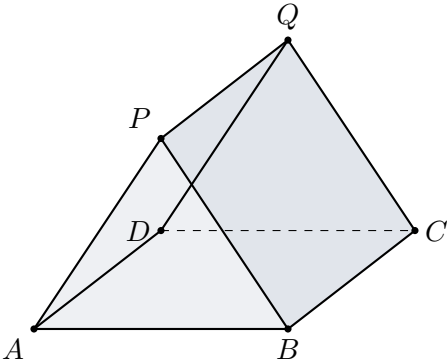
- Positive contours occur in the _____ and _____ sectors; they follow the surface's ridges.
- Negative contours occur in the _____ and _____ sectors; they follow the surface's valleys.
- Each nonzero contour has _____ components.

Fall 2025 term test problem. A triangular prism tent

The four corners of the tent are

$$\begin{aligned} A &= (0, 0, 0), & B &= (2, 0, 0), \\ C &= (2, 3, 0), & D &= (0, 3, 0), \end{aligned}$$

and the ridge endpoints are $P = (1, 0, 4)$ and $Q = (1, 3, 4)$. Let $h(x, y)$ be the height, in metres, of the tent at position (x, y) ; the ground has height 0.



Setup. Find a formula for the roof height. The height is independent of _____; it rises linearly to the ridge and then falls linearly.

Height function to complete

$$h(x, y) = \begin{cases} \text{_____}, & 0 \leq x \leq 1, \quad 0 \leq y \leq 3, \\ \text{_____}, & 1 \leq x \leq 2, \quad 0 \leq y \leq 3. \end{cases}$$

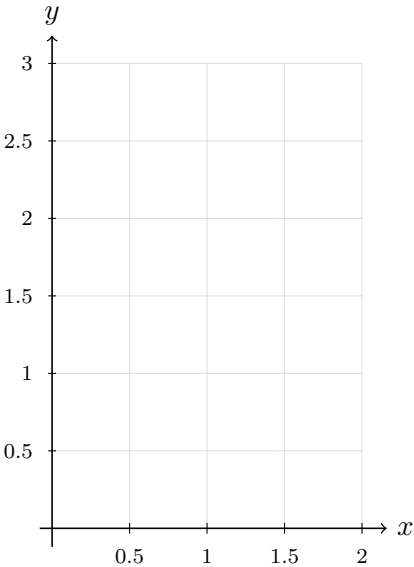
For a level $0 < c < 4$, solve $h(x, y) = c$:

$x = \text{_____}$ or $x = \text{_____}$.

Tent problem, continued. Contours and Euclidean distance

(a) Sketch at least three contours of h . Label each contour with its height.

Height c	x -location(s)
0	_____
1	_____
2	_____
3	_____
4	_____



(b) Your shoes are at $(0, 1, 0)$. A bug is on the roof at

$$\left(\frac{1}{2}, 1, h\left(\frac{1}{2}, 1\right)\right).$$

Find the Euclidean distance between the bug and your shoes.

Height of the bug: $h\left(\frac{1}{2}, 1\right) =$ _____

Bug coordinates: $\left(\text{_____, _____, _____}\right)$

Problem-solving map

Use _____ slices for cross-sections, _____ slices for contours, and missing variables to recognize _____.